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Subject Name: Database Management System

Week 1

Module 1 : Introduction to Database

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Lecture 4 –DATA abstraction, database languages, data independance

Lecture 5 –DATA model, Keys,

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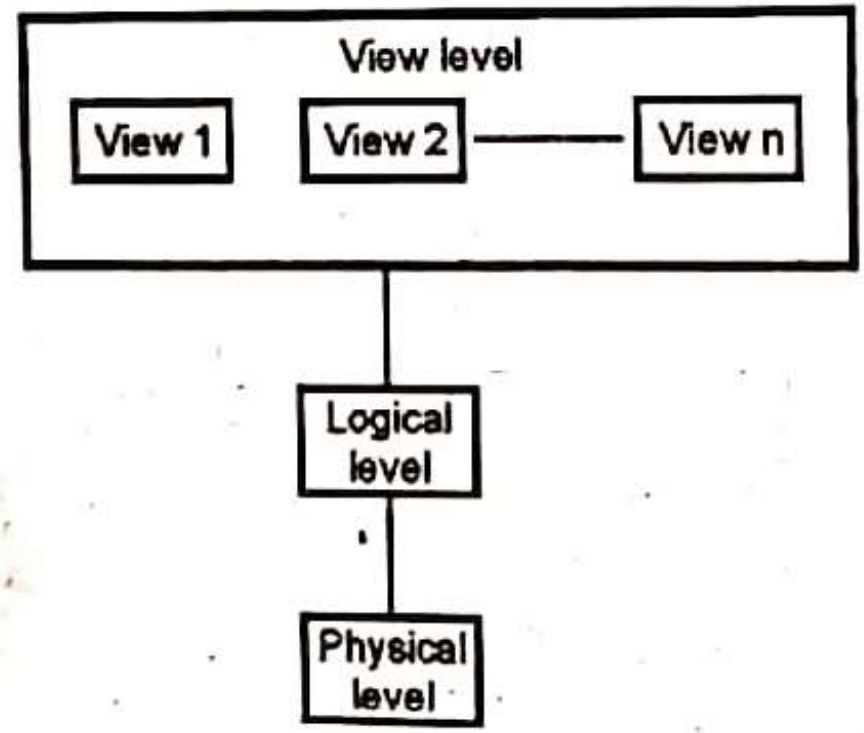
Lecture 4 –DATA abstraction, database languages, data independence



DATA ABSTRACTION:

- DATA ABSTRACTION: There are three levels of data abstraction in a database; and each level is described by a schema as explained below:

1. **Physical Level:** This is the lowest level of abstraction. At this level, a Physical Schema describes "how data is physically stored" and "what are the access methods" supported by the database. The Physical Schema will normally involve complex structures for physical storage, with the sole aim of achieving an efficient access of the database.



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2. **Logical Level or Conceptual Level**: This is the intermediate level of abstraction.

- At this level, a Logical Schema (or Conceptual Schema) describes "what data is stored in the database", "what are the data types", "what are the relationships amongst the data" and "what are the integrity constraints to be enforced on the database". This Schema is used by Database Administrators (DBA).
- The DBA decides what information is to be kept in the database. As compared to the physical level, the logical level is described by relatively less complex structures.
- The user operating at logical level need not be aware of the complexities existing at the physical level.

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3. **View Level:** This is the highest level of abstraction. At this level, there will be many Views, defined for different categories of users.

- A View for a certain group of users will describe "what subset of the database is to be made visible to that group of users". A View will describe only a subset of the underlying database, i.e. the subset which the intended group of users is authorized to access.
- There may be many Views, tailored to the specific needs of various groups of users. At the view level, the main goal is to provide an efficient and a user friendly human-interaction with the system. So, the interface at this level is made as simple and user-friendly as possible.
- A user doesn't have to be aware of the complexities at the conceptual level and at the physical level.

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- **Schema :** The overall design of the database is called schema
- For example - In a program we do variable declaration and assignment of values to the variable. The variable declaration is called **schema** and the value assigned to the variable is called **instance**.
- The schema for the student record can be RollNo, Name, Marks.
- **Instances:** When information is inserted or deleted from the database then the database gets changed. The collection of information at particular moment is called instances.
- For example - following student data base is an instance of students database

RollNo	Name	Marks
10	AAA	43
20	BBB	67

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- Types of Schema : The database has several schema based on the levels of abstraction.

1. **Physical Schema** : The physical schema is a database design described at the physical level of abstraction.
2. **Logical Schema** : The logical schema is a database design at the logical level of abstraction.
3. **Sub schema**: A database may have several views at the view level which are called sub schemas.

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DATABASE LANGUAGES

- DBMS supports two kinds of languages, one called Data Definition Language (DDL) to **specify the Database Schema** and the other called Data Manipulation Language (DML) to enable **accessing and manipulation** of the data stored in the database.

1. **DDL**: A database schema is specified by a set of definitions expressed in DDL. In a Relational Database, the result of interpretation of DDL statements is a set of tables, which are stored in the Data Dictionary or Data Directory or DBMS Catalog.

- This data stored in **Data Dictionary** is called metadata, i.e. data about data. Whenever the database is to be accessed, the DBMS will first make a reference to the Data Dictionary with a view to determine the structure of data to be accessed;

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- only then it will access the actual data in the database. Thus, the data dictionary is accessed during the processing of each and every query.
- The storage structure and access methods used by a database are specified by a set of definitions in a special type of DDL called Data Storage and Definition Language. The result of interpretation of these definitions will be a set of physical schema structures and a set of access methods supported by the system. These details are usually hidden from the database-users.

2. **DML**: A DML is a language that enables users to insert, delete, update or read data in the database. A DML query is a statement specifying information to be accessed for retrieval or insert/update/delete. The subset of DML that performs retrieval of information is called a query language.

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2. The goal of DML is to provide an efficient and user-friendly human interface to the database, for performing the following operations:

   (i) Retrieval of information from the database

   (ii) Insertion of new information into the database.

   (iii) Deletion of information from the database.

   (iv) Update of information stored in the database.

- **Procedural DML**= what data is to be extracted from DB + How to extract those data (not easier , efficient queries than non procedural)
- **Non procedural DML**= what data is to be extracted from DB

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DATA INDEPENDENCE :

- The ability of a DBMS to modify its schema at one level, without affecting the schema at the next higher level, is called data independence.
- There are two levels of data independence:

  (a) Physical Data Independence The Physical Data Independence refers to the ability of DBMS to modify physical schema, without affecting the logical schema or the views. The modifications at the physical level may be driven by advancements in hardware technology, but these modifications should not cause any changes at the logical level.

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(b) Logical Data Independence: The Logical Data Independence refers to the ability of DBMS to modify the logical schema without causing any changes at the view level.

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Lecture 5 –DATA model, Keys



Data Model

- A data model comprises a collection of conceptual tools for describing data, the data relationships, the data semantics (meaning) of data & constraints.
- Data model is structure below the database. Data model provides a way to describe the design of database at physical, logical & view level
- Thus, model defines structure of database.
- Categories of data model - There are two categories of logical data models.

a. Object based logical models

b. Record based logical models.

a. Object based logical model - The object based logical models view the universe as a collection of objects. as follows: - it's as classified into two sub categories:

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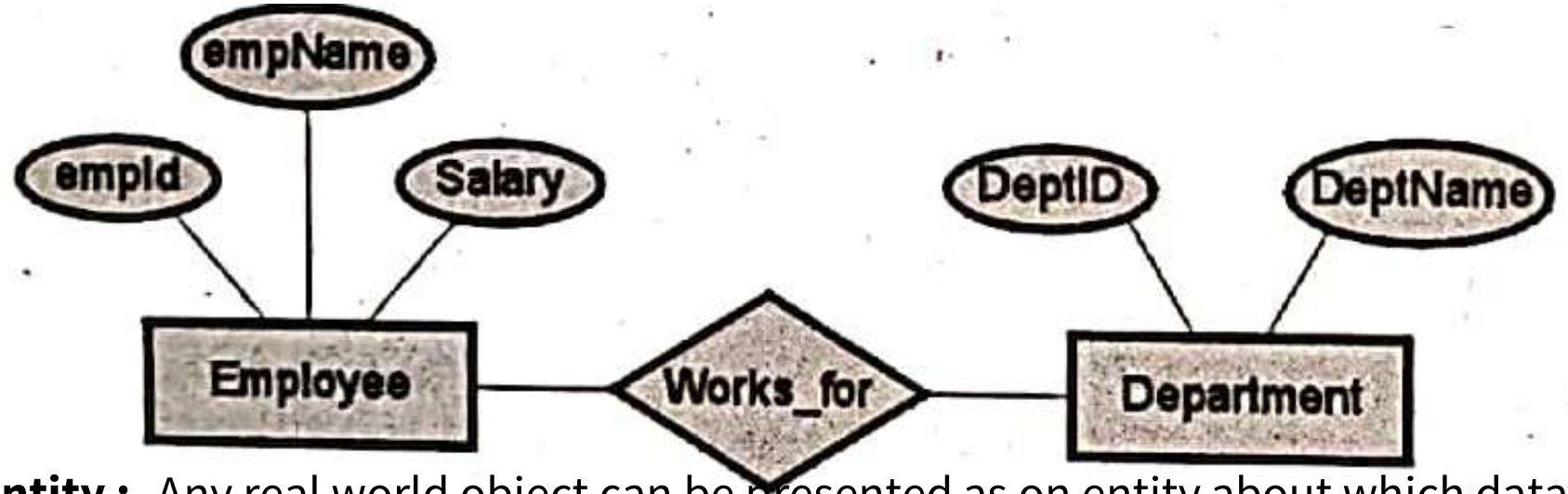
i) Entity - Relationship model

ii) Object Based Data model.

i) Entity - Relationship Model :

- As the name suggests the entity relationship model uses collection of basic objects called entities of relationships.
- The entity is a thing thing or object in the real world.
- The entire entity relationships model widely used in database design.
- eg :- following is representation of Entity relationship model in which the relationship Work for is between entities Employee & Department.

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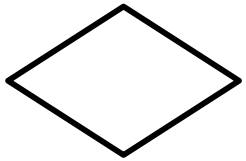


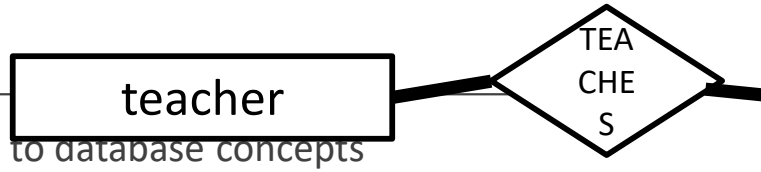
• **Entity :-** Any real world object can be presented as on entity about which data can be stored in a database. All the real world objects like book, an organization, a product, a car, a person are examples of an entity.

Symbol--→ 

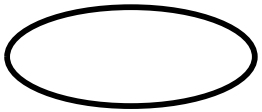
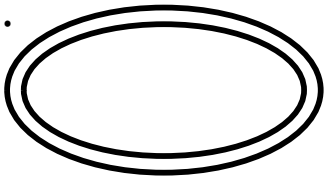
Example --→ student

• **relationship :** rhombus is used to set up relationship between two or more entities. it's defined as an association among entity sets.

Symbol--→ 

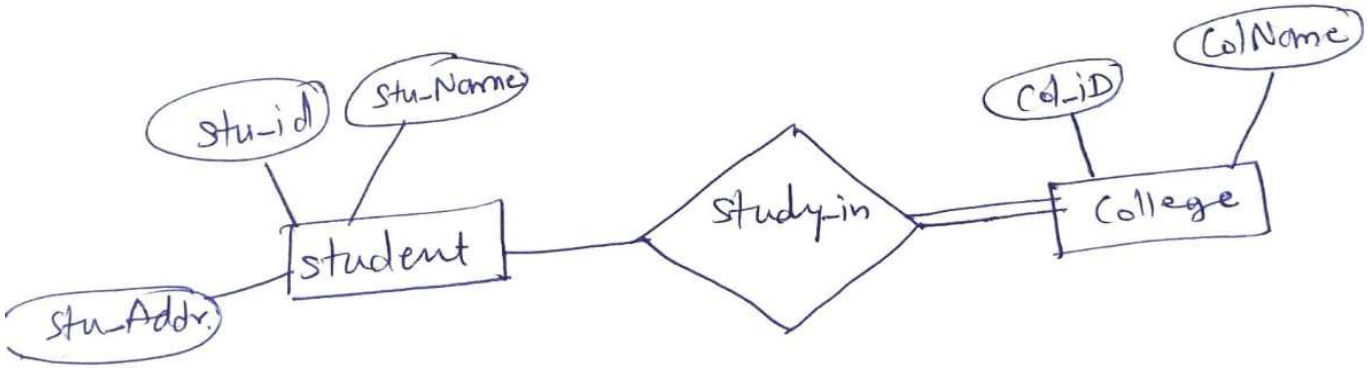
Example --→ 

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- Attributes - Each entity has set of properties. These properties of each entity are termed as attributes for example, a car entity would be described by attributes such as price, registration Number; Model No. & colour. etc.
- Symbol--→ 
- Multivalued attributes: An attribute that can hold multiple values is known as Multivalued attributes.
- Symbol--→ 
- Derived attributes: derived attributes are those which are derived based on other attributes. Eg- Age

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- Total participation: Each entity is involved in the relationship. Total Participation is represented by double lines.



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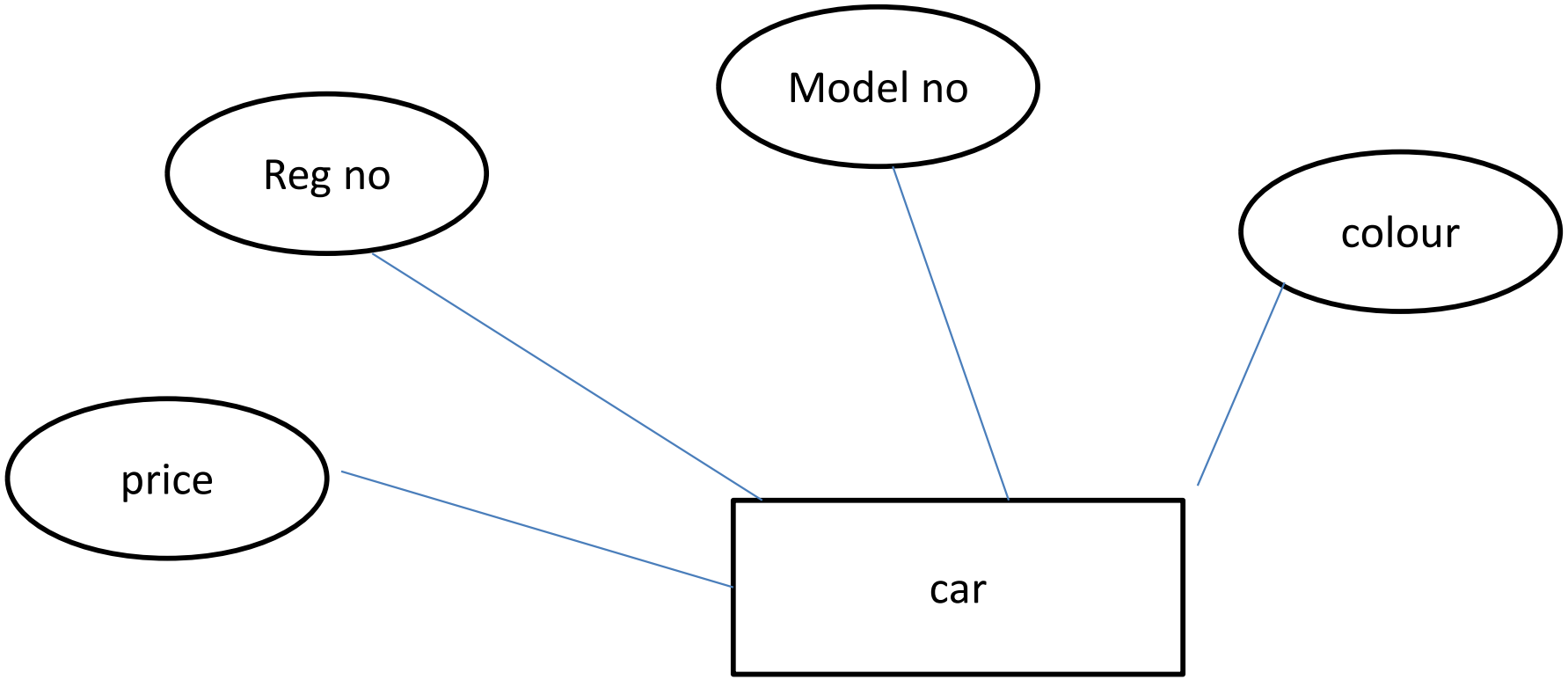
- ✓ Advantages of E-R Model :
- simple :it's easier to draw ER diagram when we know entities & relationship.
- easy to understand :- it's logical so easy to design.
- effective :- it's effective communication tool.
- Integrated: ER Model can be easily integrated with Relational model.
- Easy Conversion :- easy to convert into other type of models.

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- ✓ Disadvantages of ER Model
- loss of information :- while drawing ER diagram some information could be hidden or lost.
- limited relationships : it represents limited relationship compared to others.
- No representation for data Manipulation : it's not possible to represent data manipulation in ER Model.
- No industry standard - There is no industry standard for notations of ER diagram,

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Eg of entity and attributes:



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object-based data models -

- Like E-R model, this model also depicts database as a collection of objects as below:

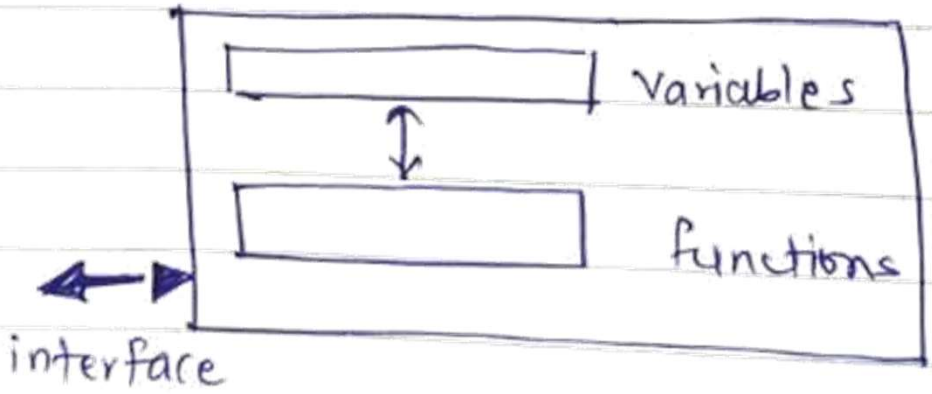


Fig. Object.

• An object Body encapsulates data (Variables) & Methods (functions) to manipulate data (variable). The objects that have same set of variables & same set of functions are grouped together as a class. Thus, a class may be viewed as a type definition of objects.

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- The only way an object "A" can access data item: inside another object (say B) is by invoking methods of "B"

- The object A can accomplish this by making B's methods, through B's interface
- In simple words we can say as, A database (object -Based) is modeled. as a set of classes & database will comprise objects belonging to those classes.

b) Record based Logical Model : These models describe data at the logical level, as a collection of fixed-format records of different types.

- Each record type can have a fixed no. of fields (Attribute). & each field is usually of fixed length.
- following are the subcategories of it.

i) Hierarchical Model

ii) Network Model iii) Relational Model.

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i) **Hierarchical Model** →The first commercial DBMS, based on this model, "Information Management system" (IMS), released by IBM in 1966.

- In this model, data is represented as records. & records are organized as a collection of trees.
- The tree structure permits only one parent record for each record. This enables modeling of only one to many relationship (not many to many relationship) amongst the records.
- The relationships among data is represented by linksor pointers.

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- Eg:-

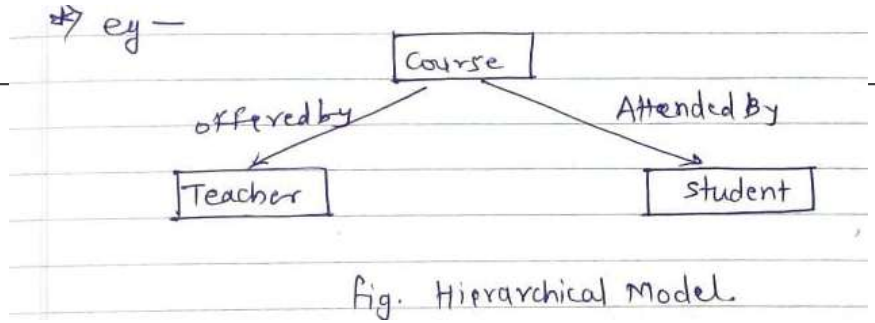


fig. Hierarchical Model.

- The above modelling does not include the relationships "what courses are being offered by a faculty", “what Courses being attended by student", "who are student taught by a faculty" & "who are the faculty members teaching a student".
- This is due to limitation of tree structure, since a node can have only one parent node. Therefore we can represent 1 to many relationships but not many to many not possible.

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ii) **The Network model:** is a model in which data is organized in nodes & links. it' resembles graph data structure.

- In this model record can be derived from more than one parent records.
- Entities are represented as a connected N/w with each other.
- The network model allows1:1 (one to one);1:M (one to many) & M to M (Many to many) relationships among entities.

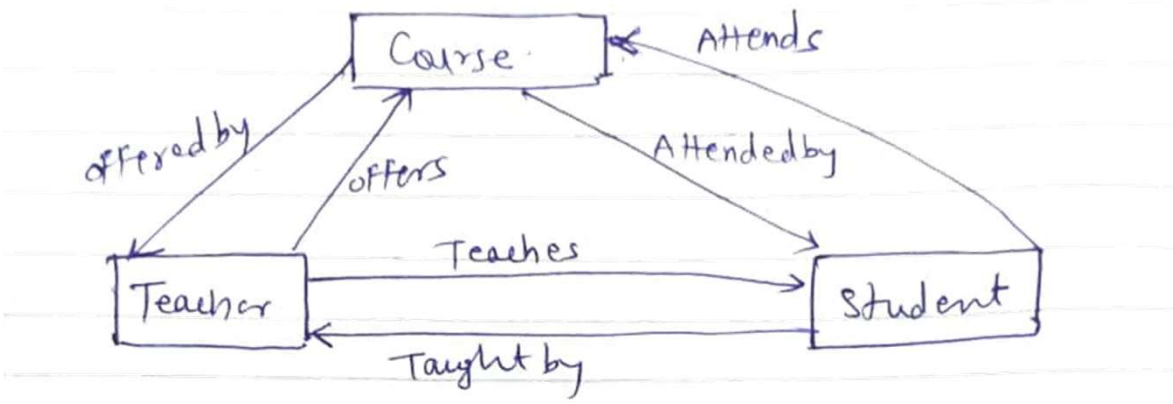


Fig: N/w model

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- ✓ Advantages:
- Simplicity Network model is conceptually simple & easy to design.
- Handles more relation types: - it can handle many to many relations. one record can be associated with the more than one other records.
- Data Access: it provides easier & flexible access. to data
- Data integrity: In Network, there is no isolated record exists in relationship with other record So, there is Data integrity or consistency of data is maintained.
- Database standard: The N/W model is based on standards.

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✓ Disadvantages:

- Complex - The overall structure become complex. & pointers of all records need to be maintained.
- Operational Anomalies - As there are many to many association among relationships; insertion deletion or updation in one record may not get reflected in other record. This is called operation Anomalies.

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iii) **Relational Model** : Relational Model consist of collection of tables which stores data & also represents the relationships among the data.

- The table is also known as relation. The table contains one or more columns & each column has unique name.
- Each table contains a record of particular type & each record type defines fixed number of fields or attributes.
- Eg: - Following figure shows the relational model which shows relationship between student & Result database. j.e. student Ram lives in city chennai & his marks are 78. Thus relationship between these two databases is maintained by the Seat No. Column.

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Advantages of Relational Database model:

- ✓ **structural Independance** - It's an ability that allows us to make changes in one database structure without affecting other. Hence making required changes in the database is convenient in relational database model.
- ✓ **Conceptual Simplicity** - The relational model allows the designer to simplify focus on logical design & not on physical design. Hence relational models are conceptually simple to understand.
- ✓ **Query capability** - Using simple Query Language [SL] user can get information from the database or designer can manipulate the database structure.
- ✓ **Easy design, maintenance & usage** - The relational can be designed logically hence they are easy to maintain & use.

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Disadvantages of Relational Database Model. -

- ✓ Relational model requires powerful Hardware & large data storage devices.
- ✓ May lead to slower processing time.
- ✓ Poorly designed systems Lead to poor implementation of database systems

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Keys

- ***Definition***: A **key** is an **attribute** or **set of attributes** that **uniquely identifies** any record (or tuple) from the table.
- ***Purpose:***
  - **Key** is used **to uniquely identify any record or row of data from the table.**
  - It is also used to establish and identify **relationships** between **tables.**



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Types of Keys:

1. Super Key
2. Candidate Key
3. Primary Key
4. Alternate Key
5. Foreign Key
6. Composite Key

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- A **super key** is a combination of all possible attributes that can uniquely identify the rows (or tuple) in the given relation.
  - Super key is a superset of a candidate key
  - A table can have many super keys
  - A super key may have additional attribute that are not needed for unique identity

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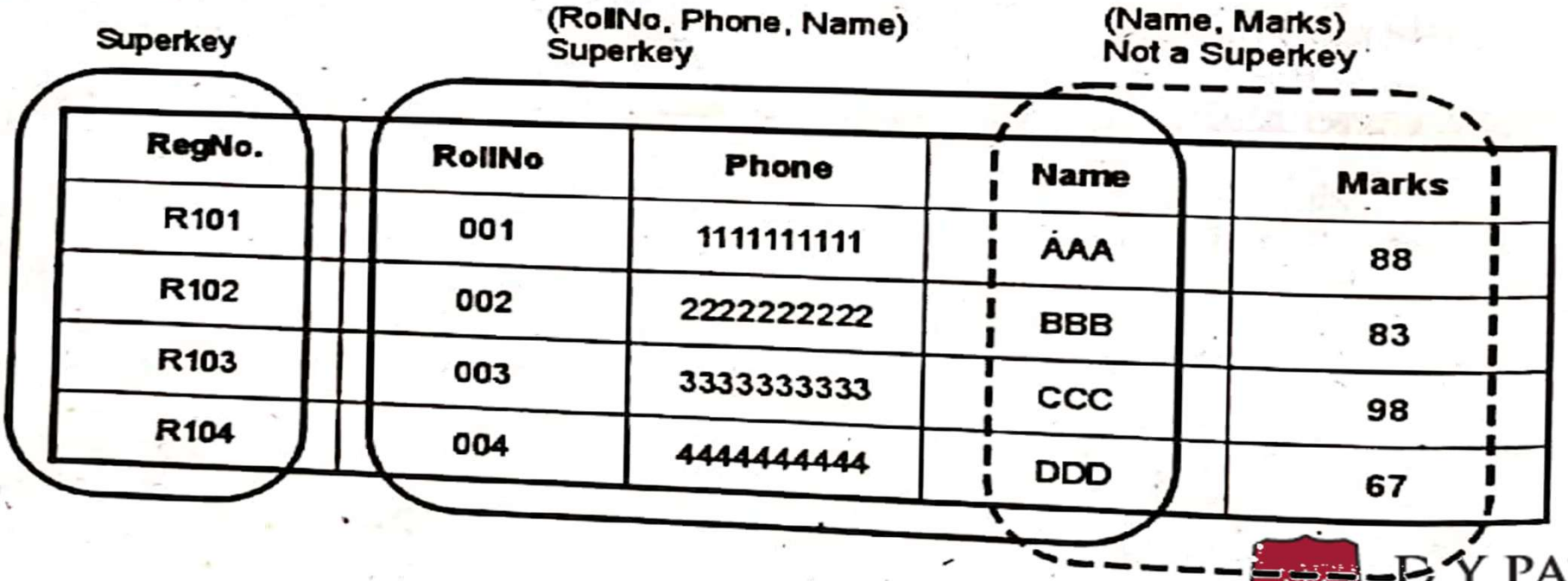
Emp_Id	Name	Aadhar_No	Email_Id	Dept_Id
01	Aman	775762540011	aa@gmail.com	1
02	Neha	876834788522	nn@gmail.com	2
03	Neha	996677898677	ss@gmail.com	2
04	Vimal	796454638800	vv@gmail.com	3

Super Keys:

1. {Emp_Id}
2. {Aadhar_No}
3. {Email_Id}
4. {Emp_Id, Aadhar_No}
5. {Aadhar_No, Email_Id}
6. {Emp_Id , Email_Id}
7. {Emp_Id, Aadhar_No, Email_Id}
8. {Emp_Id, Name}
9. {Emp_id, Name, Dep_Id}
10. {Emp_Id,Name, Aaadhar_No, Email_Id, Dept_Id}, etc....

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Reg No.	Roll No	Phone	Name	Marks
R101	001	1111111111	AAA	88
R102	002	2222222222	BBB	83
R103	003	3333333333	CCC	98
R104	004	4444444444	DDD	67



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Candidate key

- A candidate key is an attribute or set of an attribute which can uniquely identify a tuple.
- A candidate key is a **minimal super key**; or a Super key with no redundant attributes.
  - It is called a minimal super key because we select a candidate key from a set of super key such that selected candidate key is the minimum attribute required to uniquely identify the table
- Candidate keys are defined as distinct set of attributes from which **primary key can be selected**

  Candidate keys are not allowed to have NULL values



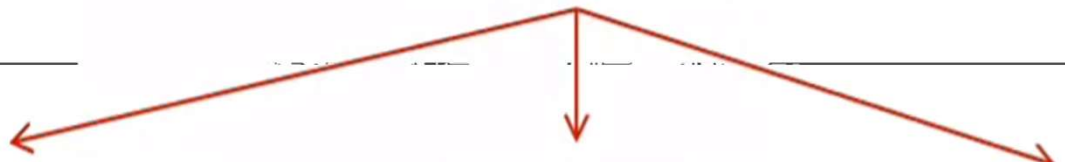
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Candidate key

Candidate key

RegNo.	RollNo	Phone	Name	Marks
R101	001	1111111111	AAA	88
R102	002	2222222222	BBB	83
R103	003	3333333333	CCC	98
R104	004	4444444444	DDD	67

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Emp_Id	Name	Aadhar_No	Email_Id	Dept_Id
01	Aman	775762540011	aa@gmail.com	1
02	Neha	876834788522	nn@gmail.com	2
03	Neha	996677898677	ss@gmail.com	2
04	Vimal	796454638800	vv@gmail.com	3

- **Candidate Keys**
  1. {Emp_Id}
  2. {Aadhar_No}
  3. {Email_Id}

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Primary Key

- A **primary key** is one of the candidate key chosen by the database designer to uniquely identify the tuple in the relation
  - The value of primary key can never be NULL.
  - The value of primary key must always be unique (not duplicate).
  - The values of primary key can never be changed i.e. no updation is possible.
  - The value of primary key must be assigned when inserting a record.
  - A relation is allowed to have only one primary key.

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Primary Key

Emp_Id	Name	Aadhar_No	Email_Id	Dept_Id
01	Aman	775762540011	aa@gmail.com	1
02	Neha	876834788522	nn@gmail.com	2
03	Neha	996677898677	ss@gmail.com	2
04	Vimal	796454638800	vv@gmail.com	3

- **Primary Key**

  1. {Emp_Id}

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Primary key

RegNo.	RollNo	Phone	Name	Marks
R101	001	1111111111	AAA	88
R102	002	2222222222	BBB	83
R103	003	3333333333	CCC	98
R104	004	4444444444	DDD	67

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Alternate Keys

- Out of all candidate keys, only one gets selected as primary key, remaining keys are known as alternate keys.
- In the Employee table
  - **Emp_Id** is best suited for the **primary key**.
  - Rest of the attributes like **Aadhar_No, Email_Id** are considered as a **alternate keys**.

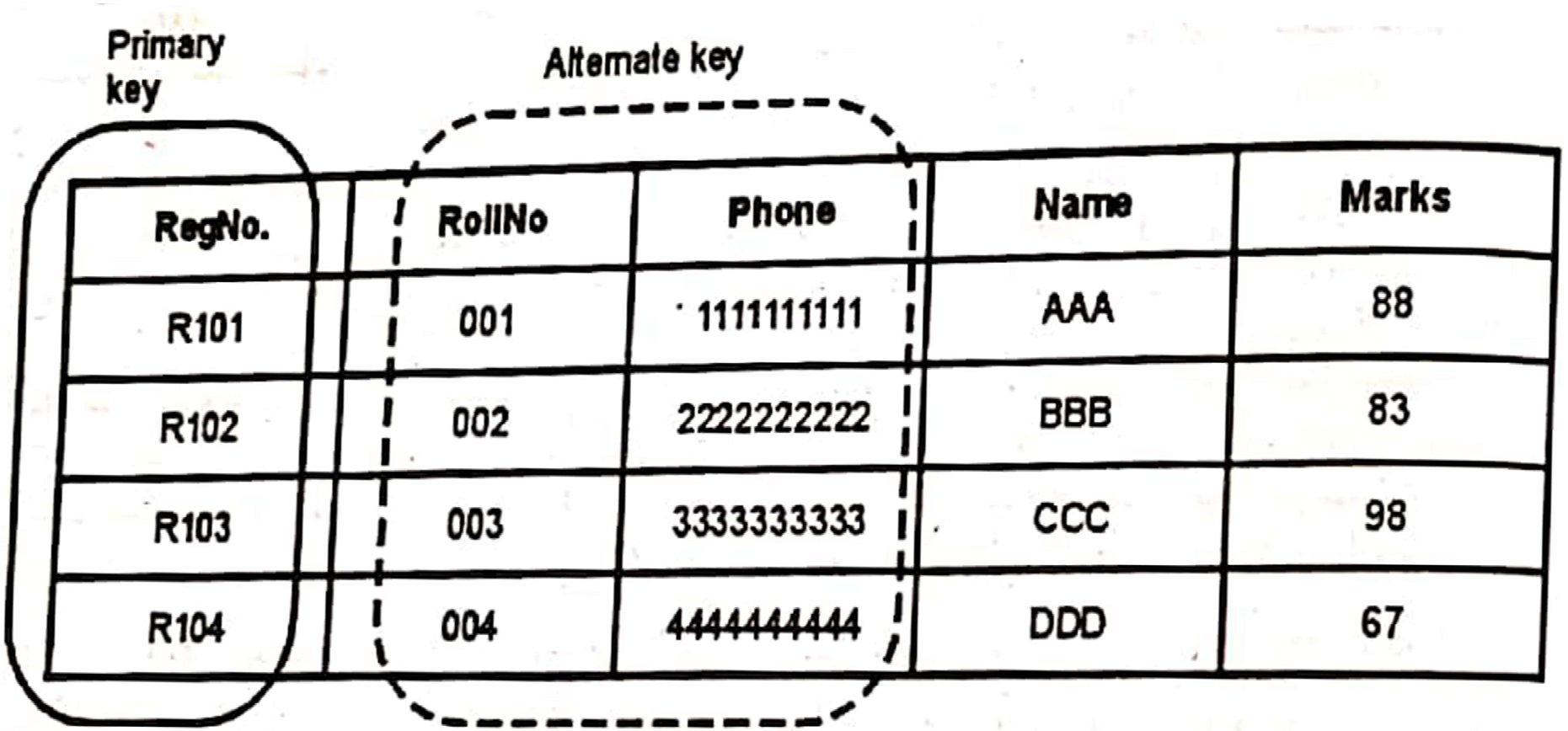
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02	Neha	876834788522	nn@gmail.com	2
03	Neha	996677898677	ss@gmail.com	2
04	Vimal	796454638800	vv@gmail.com	3

- **Alternate Keys**
  1. {Aadhar_No}
  2. {Email_Id}

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RegNo.	RollNo	Phone	Name	Marks
R101	001	1111111111	AAA	88
R102	002	2222222222	BBB	83
R103	003	3333333333	CCC	98
R104	004	4444444444	DDD	67

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Foreign key

- A **Foreign Key** is:
  - A **key** used to link two tables together.
  - An **attribute (or set of attributes)** in one table that refers to the **Primary Key** in another table.
- The **purpose** of the **foreign key** is
  - to ensure (or maintain) **referential integrity** of the data.

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- Foreign key references the primary key of the table.
- Foreign key can take only those values which are present in the primary key of the referenced relation.
- Foreign key may have a name other than that of a primary key.
- Foreign key can take the NULL value.
- There is no restriction on a foreign key to be unique.
- In fact, foreign key is not unique most of the time.
- Referenced relation may also be called as the master table or primary table.
- Referencing relation may also be called as the foreign table.

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Foreign Keys

Employee Table (Referencing relation)

Emp_Id	Name	Aadhar_No	Email_Id	Dept_Id
01	Aman	775762540011	aa@gmail.com	1
02	Neha	876834788522	nn@gmail.com	2
03	Neha	996677898677	ss@gmail.com	2
04	Vimal	796454638800	vv@gmail.com	3

Foreign Key:

In Employee Table

1. Dept_Id

Primary Key

Department Table (Referenced relation)

Dept_Id	Dept_Name
1	Sales
2	Marketing
3	HR



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Student

RegNo.	RollNo	Phone	Name	Marks
R101	001	1111111111	AAA	88
R102	002	2222222222	BBB	83
R103	003	3333333333	CCC	98
R104	004	4444444444	DDD	67

Parent Table

Primary key

Course

CourseID	CourseName	RegNo
C111	ComputerSci	R103
C112	Electrical	R101
C113	Mechanical	R104
C114	Civil	R102

Child Table

Foreign key

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Composite keys

- A key that has more than one attributes is known as composite key. It is also known as compound key.

Cust_Id	Order_Id	Product_Code	Product_Count
C01	001	P111	5
C02	012	P111	8
C02	012	P222	6
C01	001	P333	9

- **Composite Key:**

  {Cust_Id, Product_Code}



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Lecture 6 –Attributes in DBMS

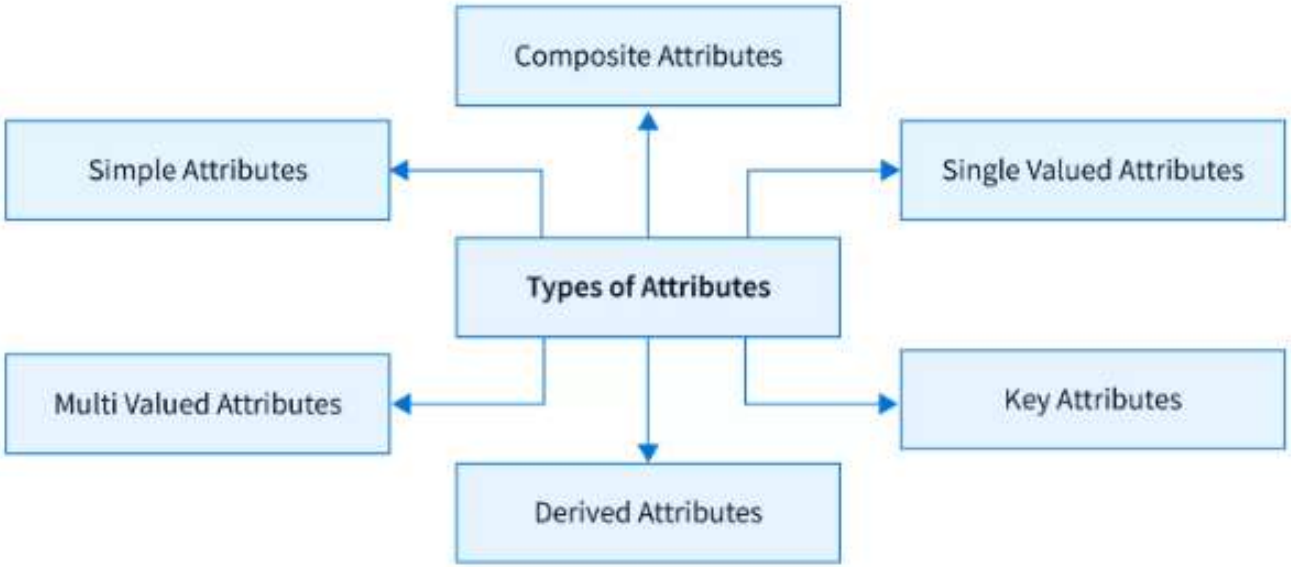


Attributes in DBMS

- In the relational database, attributes are the characteristic properties that define all the items belonging to a specific category applied to every column cell.
- We can think of attributes as the values that are used to describe a specific member or attributes as a column in an entity table.
- Suppose we have several students in a class; we can uniquely identify a student by the properties associated with a student, like a student roll number, name, branch, age, etc.
- These properties that describe an entity are attributes for students.

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Types of attributes



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student_id	student_age	student_name	student_branch
101	21	Aryan	Civil Engeneering
102	22	Eshaan	Electonics and Communication Engineering
103	24	Divyansh	Computer Science Engineering
104	21	Vaibhav Singh	Chemical Engineering

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composite attributes

- As the name suggests, those attributes composed of many other simple attributes are called **composite attributes**.
- Properties like student_name can be further divided into sub-parts like first_name, middle_name, last_name where each sub part can form an independent attributes. Similarly, address property can be composed of state, city, country etc.

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|-> Name

|-> First Name

|-> Middle Name

|-> Last Name

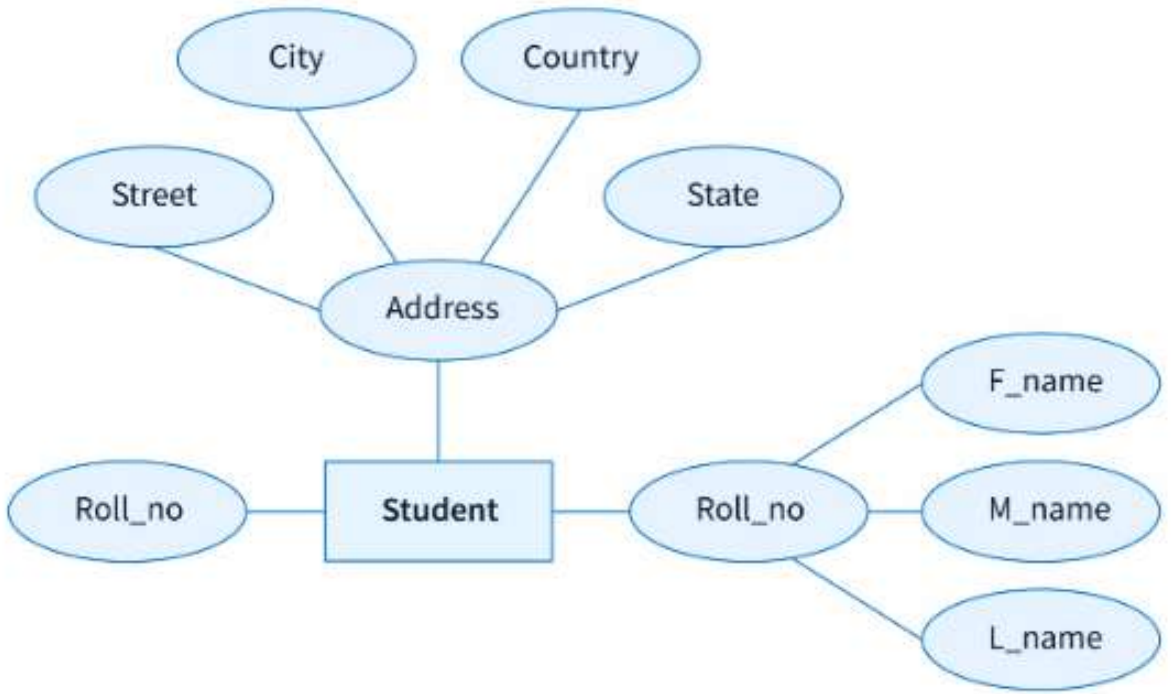
|-> Address

|-> Street

|-> City

|-> Country

|-> State



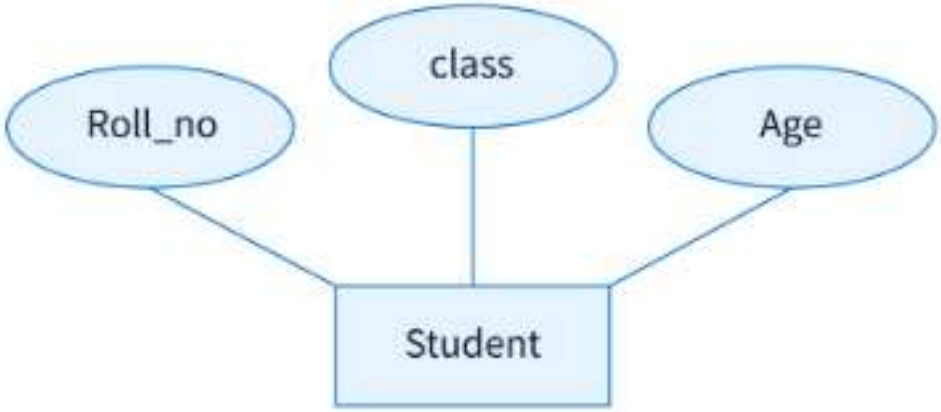
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Simple/Atomic Attribute

- Such attributes that cannot be further sub-divide into several attributes are called simple attributes.
- For example, attributes like current class and branch of a student have an absolute value and they cannot be further sub-divided into other parts like we divided full_name into first_name, middle_name and last_name.



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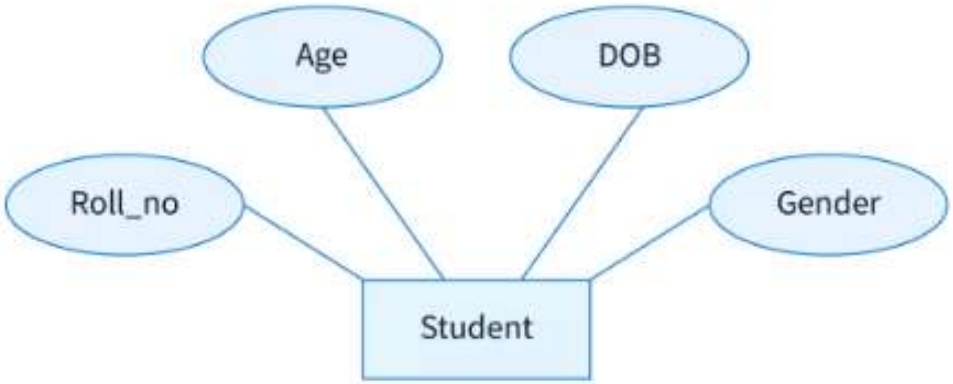
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Single-valued Attribute

- Single valued attributes are properties that have a single particular value for a particular entity. This means that single-valued attributes don't have more than one value for an item.
- For example, attributes like age, roll number, and gender associated with the student can only have a single value for an individual student.

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- **Note**: Single valued attributes are enclosed inside single-lined circles.



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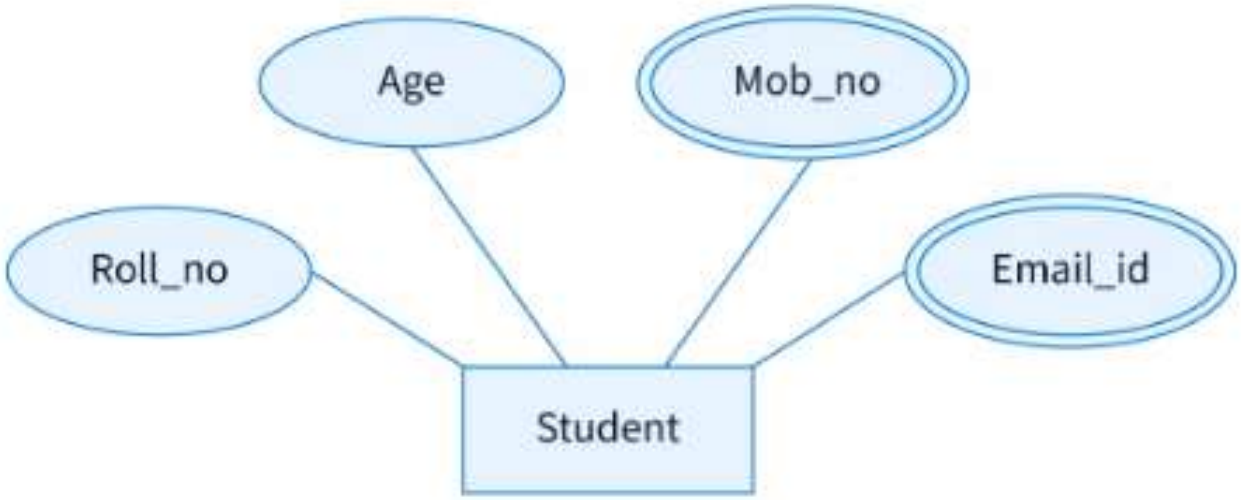
Multi-valued Attribute

- Attributes that can have multiple sets of values for a single entity are known as Multi-valued attributes. They are enclosed inside double bounded ovals in the ER diagram (ER diagram stands for **entity relation diagram** that displays the relationship of entity sets stored in a database.)
- For example, attributes like phone_number, email_id and hobbies are categorized as *multi-valued attributes* because a single student in the class can have more than one phone number, email_id and hobbies.

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- Student
- name - Eshaan
- hobbies - cycling, swimming, movies
- Here, a single student Eshaan in the database, has a set of different hobbies mentioned above and hence hobbies is a multi-valued attribute for the students.

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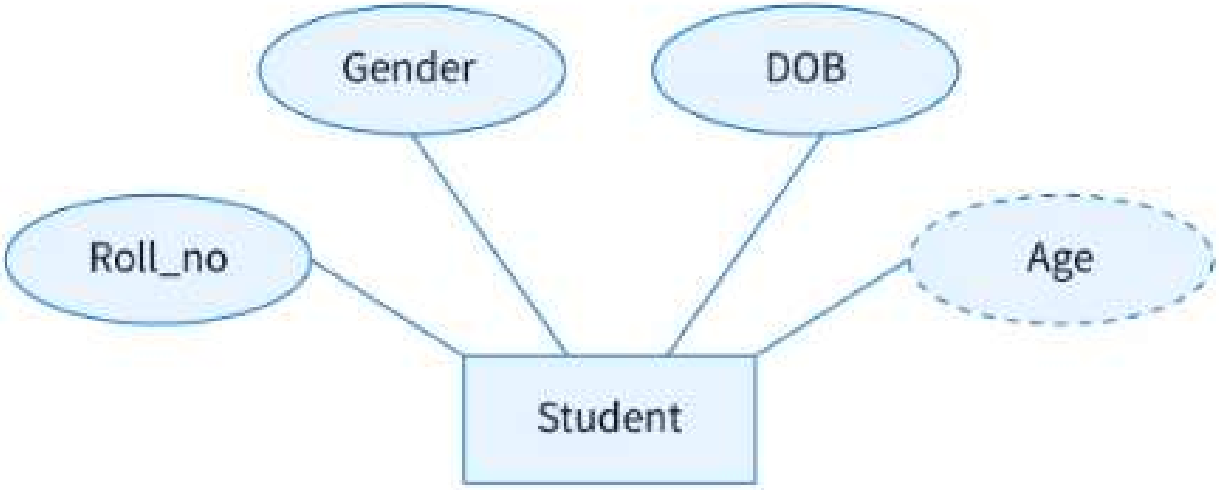
Derived/Stored Attributes

- Such attributes whose value can be derived from other attributes of the entity are called **derived attributes**.
- Valued stored in derived attributes can be obtained from processing other attributes.
- For example, properties like age of a student can easily be obtained by subtracting the current day from the date of birth attribute of the student entity. So, age is a derived attribute in the student table.



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- Age is the derived attribute.
- DOB is the stored attribute.
- **Note**: Derived attributes are marked using dotted boundaries.

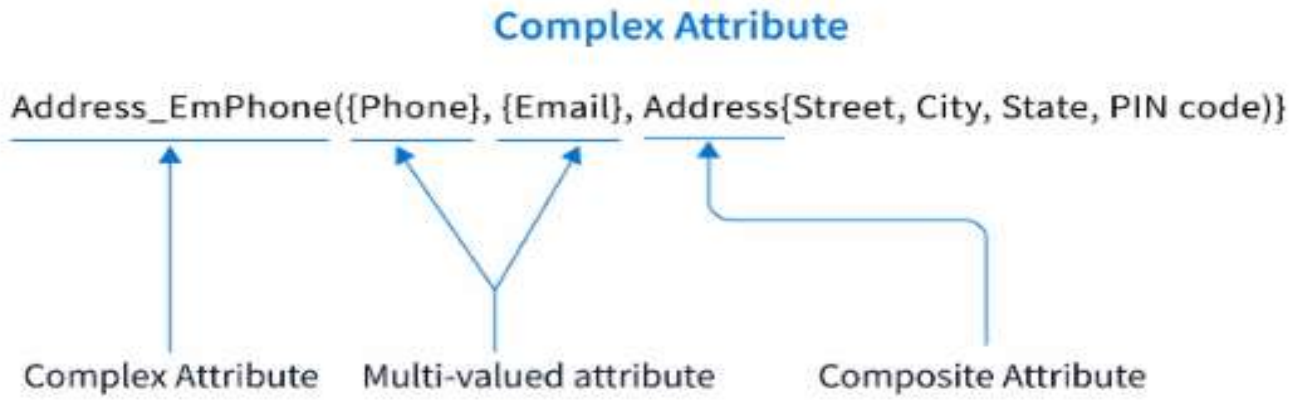


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Complex Attribute

- Nesting multi-valued and composite attributes together form a complex attribute. Complex attributes are formed by arbitrarily nesting the composite and multi-valued attributes.
- For example, a person owns more than one house, and each house has more than one phone number.
- Then, the attribute phone is represented as a complex attribute.

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- Here, in the above image, the attribute Address_EmPhone is used to store the email, phone number and address of the employee. We are enclosing multi-valued attributes inside braces {} and composite attributes inside parentheses ().

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Attributes are categorized in six different types -

- **Composite attribute**: Attribute composed of more than one simple attribute.
- **Atomic attribute**: Attributes that cannot be further sub-divide.
- **Single valued attribute**: Such attributes don't have more than one value for an item.
- **Multi-valued attribute**: These attributes have multiple set of values for a single entity.
- **Derived attribute**: Values of these attributes can be derived from other entity attributes.
- **Complex attribute**: Nesting multi-valued and composite attributes forms complex attributes.

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Thank You



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